

FarmSense AI

50 Important Viva Questions & Answers

AI-Powered Smart Farming for India

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🏠 Architecture & Design

Q1. Why does FarmSense AI use two backends — Node.js and Django?

'œ Node.js handles real-time I/O, JWT auth, Socket.io, email/SMS, and scheduling. Django handles all ML/AI tasks using Python-only libraries like scikit-learn, LangChain, and Pillow. Each server does what it is best at.

Q2. Explain the 3-tier architecture of FarmSense AI.

'œ Tier 1: React frontend (port 5173) — farmer UI. Tier 2: Node.js backend (port 5000) — API gateway, auth, scheduler, notifications. Tier 3: Django AI engine (port 8000) — ML models and AI chatbot.

Q3. How does Node.js communicate with Django securely?

'œ Node.js sends all requests to Django with an X-Internal-Key header. Django rejects any request that does not include this key. This prevents external users from calling the AI engine directly.

Q4. Why is Django never exposed directly to the internet?

'œ Because Django only runs ML models — it has no authentication system for external users. All security (JWT, CORS, rate limiting) is handled by Node.js which acts as a gateway.

Q5. What is the role of Socket.io in FarmSense AI?

'œ Socket.io provides real-time WebSocket connections. When a critical alert is generated, the server emits a new_alert event to the specific farmer's browser room (farm_{id}) so alerts appear instantly without refreshing.

🧠 AI & Machine Learning

Q6. Name all 5 ML models used in FarmSense AI.

'œ 1. Crop Recommendation — NPK + weather !' best crop. 2. Fertilizer Optimizer — soil + crop !' fertilizer plan. 3. Irrigation Advisor — weather + crop stage !' water needs. 4. Disease Detection — image !' diagnosis. 5. Yield Predictor — farm data !' expected yield.

Q7. What algorithm is used for Crop Recommendation?

'œ A scikit-learn classifier (e.g., Random Forest or Decision Tree) trained on a dataset of soil NPK, pH, temperature, humidity, and rainfall to predict the most suitable crop.

Q8. What is RAG and how is it used in FarmSense AI?

'œ RAG = Retrieval-Augmented Generation. Farming knowledge is stored in ChromaDB as vector embeddings. When a farmer asks a question, relevant knowledge chunks are retrieved and passed to the Groq LLM along with the question, producing a grounded, accurate answer.

Q9. What is ChromaDB?

'œ ChromaDB is a vector database that stores text as numerical embeddings for semantic similarity search. FarmSense uses it as the knowledge base for the RAG chatbot.

Q10. Why was Ollama replaced by Groq API?

'œ Ollama runs LLMs locally and requires a GPU. When local setup failed, Groq API was used — it runs the same open-source models (Llama 3.1, Qwen) on cloud LPU hardware, providing fast inference without local GPU requirements.

Q11. What is the Full Pipeline endpoint?

'œ A single Django endpoint (/api/analysis/full-pipeline/) that runs all 5 ML models in one call. It accepts farm data and returns crop recommendation, fertilizer plan, irrigation advice, and yield prediction together.

Q12. What model is used for Disease Detection?

'œ Groq Vision API using the qwen/qwen3.6-27b model. The crop image is sent to this model which returns the disease name, severity, treatment, and prevention advice in structured JSON.

Q13. What is an embedding in the context of RAG?

'œ An embedding is a numerical vector that represents the semantic meaning of text. Similar texts produce similar vectors, enabling ChromaDB to find relevant knowledge chunks based on meaning, not just keyword matching.

Q14. What Python libraries power the AI engine?

'œ scikit-learn (ML models), LangChain (RAG pipeline), ChromaDB (vector store), Pillow (image processing), pandas/numpy (data handling), and langchain_groq (Groq API integration).

Q15. How does the Crop Comparison feature work?

'œ The farmer selects 2–3 crops. The system runs the ML model for each crop using the farm's soil and location data, then returns a comparison of suitability score, expected yield, water needs, and market price.

&j Node.js Backend

Q16. What is JWT and how is it used in FarmSense AI?

'œ JWT (JSON Web Token) is a stateless authentication method. On login, the server generates a signed token. The client sends it in every request header. The auth middleware verifies the token without hitting the database on each request.

Q17. What does the node-cron scheduler do?

'œ 6 AM: Fetch weather for all farms !' generate alerts !' save to DB !' send Email + SMS. 7 AM: Run full AI pipeline for all farms !' store results. 8 AM: Send daily summary emails to all farmers.

Q18. What is Helmet.js?

'œ Helmet.js is a Node.js middleware that sets secure HTTP response headers, protecting against common web vulnerabilities like XSS, clickjacking, and MIME-type sniffing.

Q19. How does the email notification system work?

'œ Nodemailer sends HTML emails via Gmail SMTP using an App Password. There are two email services: emailService.js for transactional emails (verification, invoice) and notifier.js for weather alert emails.

Q20. How is the SMS notification sent?

'œ Using the Twilio SDK. The notifier.js service sends SMS and WhatsApp messages to the farmer's registered phone number when critical weather alerts are generated.

Q21. How is the PDF invoice generated without saving a file?

'œ PDFKit generates the PDF in memory as a Node.js Buffer. The buffer is then directly attached to the Nodemailer email as an attachment. No file is written to disk.

Q22. What is express-validator used for?

'œ Input validation and sanitization. Every API route that accepts user input uses express-validator to check for required fields, data types, and valid ranges before processing.

Q23. How many API routes does the Node.js server have?

'œ 12 routes: /auth, /farm, /fields, /alerts, /suggestions, /analysis, /crops, /weather, /notifications, /disease, /admin, /inspections.

Q24. What is CORS and why is it configured?

'œ CORS (Cross-Origin Resource Sharing) controls which origins can call the API. FarmSense restricts it to localhost:5173 (React dev server) to prevent unauthorized domains from making API calls.

Q25. How are weather alerts deduplicated?

'œ Before inserting an alert, the server checks if the same alert_type already exists for that farm today. If it does, the insert is skipped. This prevents duplicate alerts from appearing each time the weather page is visited.

Ø<ß" React Frontend

Q26. What is Vite and why was it chosen over Create React App?

'œ Vite is a modern frontend build tool that uses ES modules for instant hot module replacement. It is significantly faster than Create React App for development builds and startup.

Q27. How is multilingual support implemented?

'œ Using react-i18next. Translation JSON files exist for English, Hindi (hi), and Gujarati (gu). The t() function wraps all UI text. Calling i18n.changeLanguage() switches the entire UI instantly.

Q28. How does the Dashboard know how many fields the farm has?

'œ The GET /api/farm route returns both the farm object and a stats object at the top level. The stats.total_fields value is stored separately in state and displayed on the Dashboard.

Q29. What is the AuthContext used for?

'œ AuthContext is a React Context that stores the logged-in user object, JWT token, and login/logout functions. It wraps the entire app so any component can access auth state without prop drilling.

Q30. What pages exist in the FarmSense frontend?

'œ Landing, Login, Register, ForgotPassword, ResetPassword, Onboarding, Dashboard, Weather, Alerts, Suggestions, DiseaseDetection, CropComparison, FarmProfile, GovSchemes, Settings, AdminDashboard.

Database

Q31. Why was PostgreSQL chosen over MongoDB?

'œ Farm data is highly relational — farms have fields, fields have crops, alerts belong to farms, suggestions link to users. PostgreSQL's JOIN queries and foreign keys handle these relationships cleanly. MongoDB is better for unstructured data.

Q32. Name the main database tables in FarmSense AI.

'œ users, farms, fields, alerts, notifications, suggestions, soil_inspections, analysis_results, notification_preferences.

Q33. How is email verification handled in the database?

'œ On registration, a random token is stored in the users table with an expiry time. The verification email contains a link with this token. When clicked, the token is validated and the user's is_verified flag is set to true.

Q34. How are password resets handled securely?

'œ A reset token is generated and stored hashed in the database with a 1-hour expiry. The email link contains the raw token. On submission, the raw token is hashed and compared. The password is stored as a bcrypt hash.

Q35. What is the relationship between farms and fields?

'œ One farm can have many fields (one-to-many). Each field has its own crop, sow date, size, and stage. The farm table stores the overall farm profile while fields table stores individual plot details.

Weather & Alerts

Q36. What weather API is used and why?

'œ Open-Meteo API. It is completely free, requires no API key, provides 16-day forecasts, and returns data for any GPS coordinates — ideal for rural Indian farm locations.

Q37. What types of weather alerts does FarmSense generate?

'œ 6 types: Heavy Rain (>50mm), Heatwave (>42°C), Fungal Disease Risk (humidity >85%, temp 22–32°C), Strong Wind (>40 km/h), Good Sowing Window (ideal conditions), Drought Risk (no rain for 10+ days).

Q38. How is the weather alert flow triggered automatically?

'œ The node-cron scheduler runs at 6 AM daily. It fetches weather from Open-Meteo for every farm in the database, runs alert generation logic, saves new alerts to PostgreSQL, and sends email + SMS notifications.

Q39. What is ET0 and how is it used?

'œ ET0 (Reference Evapotranspiration) is a measure of water loss from soil and plants. FarmSense retrieves it from Open-Meteo and uses it in the Irrigation Advisor model to calculate how much water the crop needs.

Q40. How are weather alerts shown in real-time on the browser?

'œ When a critical alert is generated, Node.js emits a new `_alert` Socket.io event to the farm's room. The React frontend listens for this event and displays a popup notification without any page refresh.

Ø=ÜÜ General Computer Science

Q41. What is the difference between REST API and WebSocket?

'œ REST API is stateless request-response: client asks, server answers, connection closes. WebSocket is a persistent bidirectional connection: either side can send data anytime. FarmSense uses REST for data fetching and WebSocket for real-time alerts.

Q42. What is a .pkl file?

'œ A .pkl file (Python pickle) is a serialized Python object. Trained scikit-learn ML models are saved as .pkl files so they can be loaded quickly in Django without retraining every time.

Q43. What is bcrypt?

'œ bcrypt is a password hashing algorithm. It adds a random salt to the password before hashing, making it resistant to rainbow table attacks. FarmSense uses bcrypt to store all user passwords.

Q44. What is the difference between Groq and Grok?

'œ Groq (without the "k") is a hardware + API company that runs open-source LLMs (Llama, Qwen) on ultra-fast LPU chips. Grok is xAI's proprietary chatbot by Elon Musk. FarmSense uses Groq API.

Q45. What is an App Password for Gmail?

'œ A 16-digit password generated by Google specifically for third-party apps. It allows Nodemailer to send emails via Gmail SMTP without using your real Google password or enabling less secure apps.

Q46. What is CORS?

'œ Cross-Origin Resource Sharing. A browser security mechanism that blocks requests from one origin to another unless the server explicitly allows it. FarmSense restricts API calls to only the React dev server origin.

Q47. What is the purpose of the .env file?

'œ Environment files store sensitive configuration (API keys, passwords, database credentials) outside of source code. Each server (Node.js, Django) has its own .env file and these are never committed to Git.

Q48. What is middleware in Express.js?

'œ Middleware is a function that runs between receiving a request and sending a response. FarmSense uses middleware for JWT authentication (auth), input validation (express-validator), CORS, and error handling.

Q49. What is the purpose of the X-Internal-Key header?

'œ It is a shared secret between Node.js and Django. Django checks for this key in every incoming request and rejects any request that does not have it. This ensures only Node.js can call the AI engine.

Q50. What is hot module replacement (HMR)?

'œ A feature of Vite where code changes in the editor are reflected in the browser instantly without a full page reload. This speeds up frontend development significantly.